

A Lean 4 formalisation of “ $\zeta(5)$ is irrational” (A. Fauzan, 17 Sep 2026): what it proves and what it assumes

Companion note to docs/STATUS.md

23 September 2026

Abstract

The project compiles. There is one theorem, `Zeta5.zeta5_irrational : Irrational zeta5`, and the paper’s entire route to it — Theorem 2.1, Propositions 2.2, 4.1, 4.3, 5.1, 5.2, 6.3, Lemma 3.3 and the numbered equations between them — is wired up beneath it with nothing short-circuited. It rests on *two* external axioms (Hermite’s integral formula for the Hurwitz zeta function, and the prime number theorem in the partial-summation form the proof consumes; Mathlib has neither) and on *one* named **sorry**: the paper’s (6.14), the logarithmic-energy bound of §6. The four other large steps — Lemma 3.3 at the entries of (3.11), the inner-range local analysis behind Proposition 4.1, the outer-range local analysis (4.10)–(4.12), and the uniformity statement (5.7) with its §5.2 companions — are *proved*, with statements fixed before the proofs were written and no further axiom. Every arithmetic statement that the proof of Theorem 1.1 uses (§§2–5 and Appendix B) is therefore machine-checked. At three places the formal proof takes a different route from the paper’s own proof: Lemma 3.2 (the distribution formula, with the far poles), the polynomial part of the proof of Lemma 3.3, and the unimodularity arguments are replaced by different arguments, so the paper’s proofs of those three steps are bypassed rather than checked. The assumption list is recomputed from the Lean environment on every build and independently confirmed by a separate scan; §6 records the faithfulness audit against the PDF, including the places where the formal proof takes a different route from the paper’s.

1 The build

Lean 4.34.0 with Mathlib v4.34.0. Thirty-two modules plus the root file, about 24 500 lines of Lean. With all project build artifacts removed, a rebuild of every `Zeta5` module from source takes about 4–5 minutes and ends with

Build completed successfully (8957 jobs).

with zero errors and exactly *one* declaration uses ‘sorry’ warning, for `RealBound.eq_6_14` — the same one the dependency walker reports below. The other ≈ 248 warnings are Mathlib deprecation and style lints.

2 The one theorem and its assumptions

<code>theorem Zeta5.zeta5_irrational : Irrational zeta5</code>
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with `zeta5 : ℝ := ∑' v : ℕ, 1 / ((v : ℝ) + 1) ^ 5`, i.e. $\zeta(5)$. `Zeta5/Audit.lean` prints, on every build:

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ASSUMPTION REPORT for Zeta5.zeta5_irrational

(A) standard Lean axioms:
    Classical.choice
    Quot.sound
    propext

(B) explicit external axioms of this project (Zeta5/Axioms.lean):
    Zeta5.Axioms.hermite_pole_integral
    Zeta5.Axioms.pnt_prime_riemann_sum

(C) sorry(s) - unfinished steps of the paper's own argument, 1 in all:
    Zeta5.RealBound.eq_6_14

(D) any other axiom (must be empty):
    (none)

[self-check: walker agrees with Lean.collectAxioms, 6 axiom(s)]

every sorry of the Zeta5 namespace is used by Zeta5.zeta5_irrational.

```

The last line makes (C) a *complete* list rather than a selection. An independent scan (`cert/C3Scan.lean`, sharing no code with `Audit.lean`) of all 3272 declarations defined in the 33 `Zeta5` modules finds exactly two axioms and exactly one declaration mentioning `sorryAx`, namely `eq_6_14`.

The ground rules

Anything internal to Fauzan's argument is proved or a `sorry`, never an axiom. Neither axiom mentions any constant or table of the paper, and the only object of the paper that either one mentions is the explicit weight w of (2.10). A finite explicit computation is never an axiom: Appendix B's exact rational integrals and Appendix A's Table-1 evaluations and Table-2 tiling are *proved*. And no statement is changed to make it provable: the statements of the interface theorems were fixed before their proofs were written, and a comparison at the level of kernel terms with a snapshot taken before the last of them were proved shows that no statement or definition changed (`docs/CERTIFICATION.md`, §5).

3 (B) The two external axioms

`Zeta5/Axioms.lean` is the only file containing axiom declarations, and it contains two. Each docstring gives the statement, a citation, and why Mathlib lacks it.

`hermite_pole_integral`

$$\int_0^\infty \frac{w(y)}{y^2 + a^2} dy = a^4 \zeta(5, a) - \frac{1}{2a} - \frac{1}{4} \quad (a > 0),$$

with w the weight of (2.10) and $\zeta(5, a) = \sum_{k \geq 0} (k+a)^{-5}$: the display in the middle of p. 5. **Citation:** DLMF 25.11.29, Hermite’s integral formula for the Hurwitz zeta function — the reference the paper itself gives. **Why not Mathlib:** Mathlib has `HurwitzZeta.hurwitzZeta` and its functional equation, but no Hermite/Abel–Plana integral representation and no theory of the Eisenstein-type weight w . It is slightly stronger than DLMF 25.11.29 alone: the paper’s four integrations by parts on p. 5 are absorbed into it. Numerically confirmed from the Lean definition of w at 40 digits.

`pnt_prime_riemann_sum`

For $0 \leq a < b$ and φ bounded on $[a, b]$ and continuous off a finite set,

$$\frac{1}{X} \sum_{\substack{p \text{ prime} \\ aX < p \leq bX}} \varphi(p/X) \log p \longrightarrow \int_a^b \varphi(y) dy \quad (X \rightarrow \infty).$$

Citation: the prime number theorem in the form $\theta(x) \sim x$ (the paper’s [9, §27.12]), carried through Abel summation and the Darboux criterion (Apostol, *Introduction to Analytic Number Theory*, Thms 4.2, 4.4, §4.3). **Why not Mathlib:** as of v4.34.0 Mathlib has Chebyshev-type bounds but no $\theta(x) \sim x$. It is the *partial-summation corollary* of the PNT, the one place where a reader must grant more than a bare citation; its hypotheses carry no information about the paper’s functions, and all of them are discharged inside `PrimeSum.lean`. The certification derives from it in Lean, with no `sorry`, the dyadic form $(\theta(2X) - \theta(X))/X \rightarrow 1$ of the prime number theorem, so it is not vacuous.

4 (C) The one unfinished step

<code>sorry</code>	paper statement, and what is missing	effort
<code>RealBound.eq_6_14</code>	(6.14) , p. 20: for $\Delta_K(\zeta(5)) > 0$, $\log \Delta_K(\zeta(5)) \leq 2h(h + 6N - K) \log K + (\lambda M_0 - I(\rho))K^2 + 18h \log K + 160h$. Missing, in three separable pieces: (i) Andréief’s identity (6.10), the h -fold Gram integral, with (6.12)–(6.13); (ii) Lemma 6.2 , the zero-mass logarithmic energy $I(\nu) \leq 0$ (p. 18; Ransford, Thm 3.1.2), its application to circle-regularised configurations, and (6.6)–(6.9); (iii) (6.2), the arcsine-potential bound $2U^\rho(t) - V(t) \leq M_0$ on $[0, 2]$, which is Appendix A’s 684 certified evaluations of $\mathcal{B}(l, r)$ (A.9), together with the identification of the Lean constant <code>Irho</code> (the closed form (A.2)) with the energy $I(\rho)$.	several months

(6.14) is the one genuinely large item left, needing measure and potential theory that Mathlib has only in part and some 23 000 certified `arctan`/`log` evaluations. It holds, with large slack, at the exact values of $\Delta_K(\zeta(5))$ for $K = 40, 80, 120$ computed in the referee audit of the preprint; that is a consistency check, not a proof.

5 How the four large interface statements were proved

All four are closed in the strict sense: `#print axioms` on each shows no `sorryAx` (verbatim: `'Zeta5.outer_local_analysis'` depends on axioms: `[propext, Classical.choice, Quot.sound]`, and likewise for the other three). Nothing was moved one level down.

statement	how it was proved
shared lemma HermiteBasis. det_coeffMatrix_ unimodular	<i>A basis of integer polynomials that reduces mod p to a Hermite-interpolation basis $\prod_{c \neq a} (t + u_c)^{m_c} (t + u_a)^i$ with distinct nodes is $\mathbb{Z}_p$-unimodular:</i> linear independence over any field via coprimality of the class factors, a dimension count, and reduction of the determinant mod p . This is p. 10's "triangular local bases whose diagonal entries are units" and p. 12's "these form a unimodular basis", by one argument for both.
outer_local_ analysis, §4.2, (4.10)–(4.12)	<code>OuterBasis.lean</code> (1943 lines), following pp. 11–12 step by step: square classes and the counts $\ell - \delta$ and $\ell - 2$; the rows $P_a q_{a,i}$ of (4.11) and the zero-class rows; $v_p(\mu(t^e)) \geq -1$, with integrality for $e < 2p - 3$; the splitting $G_K = A + p^{-1}L$; $L = UL_0U^T$ with $L_0[i, j] = 0$ for $i + j < K - 6N + 2p - 3$ in the monomial basis, hence $\text{rank } L \leq r_p$ — p. 12's reason, applied in the basis where it holds; and the four entry-bound cases behind (4.12).
PrimeSum. eq_5.7_ uniformity, (5.7) and §5.2	<code>Uniformity.lean</code> (1396 lines), with the explicit constant $C = 400M^2$ (the true size is about $4M$). Inner: each class's contribution is the (5.4) integrand at $z = a/p$, and an elementary Riemann-sum estimate with two breakpoint cells bounds the error; the extras sum exactly; p. 14's transition remark is the identity $I(T+1) - I(T) = T - x - 2$. $v_p(S_K)$: Legendre with $p^2 > 2K$. Outer: $K T(p/K)$ is an explicit integer and the comparison is linear integer arithmetic in each range.
CrudeBound. crude_entry_ bound, Lemma 3.3 at (3.11)	<code>Lemma33.lean</code> (1155 lines) proves Lemma 3.3 itself in the paper's form, and the pullback (3.1) for every $\mu_X(B/D_{\text{tail}})$. Pole terms as in the paper; the polynomial part by a different route (§6, item 5).
Section3. entry_bounds_ 4.2.4.3, Prop. 4.1's basis and entries	<code>InnerTate</code> , <code>InnerGeneral</code> , <code>InnerEntries</code> (2240 lines). B is the Gram matrix in the basis (4.5), $c = (\det U)^{-2}$ a p -adic unit by the shared lemma; the entry bound $\min((4.3), \min_c(4.2))$ comes from a general local bound $v_p^G(\tau_X^{\text{ext}}(W/Q)) \geq \min_a(u_a - t_a) - 4$, proved from Raabe's multiplication theorem for τ ($m^4\tau(P) = \sum_{a < m} \tau(P(a + mz))$), via Mathlib's <code>sum_bernoulli</code>) and a per-class estimate using a <i>truncated</i> inverse of the far-pole factor.

Every sub-statement was tested in exact arithmetic before it was proved (scripts in `numerics/`): the (4.11) decomposition at $K = 40, 80, 120$ (0 failures; $\text{rank } L = r_p$ wherever computed; a must-fail control without the $+1/(2j)$ of (2.3) fails 13 times), the inner entry bounds on all 703 entries at $K = 40$ and all 2775 at $K = 80$, Lemma 3.3 on all 703 entries of (3.11) at $K = 40$ for eleven primes, and (5.7) over every prime of the ranges at K up to 520 000.

Consequences. Propositions 4.1, 4.3, 5.1, 5.2 and (3.12) are fully proved (5.2 modulo the PNT axiom). The §4.1 combinatorics (`Section41.lean`, including the repair of the gap in the p. 10 inequality, item 18 below) and the §4.2 local lemmas (`OuterLocal.lean`) are load-bearing: 104/148 and 98/107 of their declarations are in the dependency cone of the main theorem.

6 Faithfulness

Statements. The statements of the four interface theorems proved last, of `eq_6_14` and of `zeta5_irrational` are byte-identical to those recorded in a snapshot taken before the four proofs were written, and a comparison at the level of kernel terms shows that no statement or definition of that snapshot changed (`docs/CERTIFICATION.md`, §5).

Statements checked against the PDF page images (pp. 5–15, 18–20). `Lemma33.lemma_3_3` is Lemma 3.3 as printed on p. 8, for every p, K, d ; its hypothesis “ A integer-valued on $\mathbb{Z}_p$ ” is taken as $v_p(A(z)) \geq 0$ for $z \in \mathbb{Z}$, which is equivalent by density. `pullback` is (3.1) on the whole domain of μ_X . `raabe_tau` is (3.7) for polynomials. The weights of `OuterBasis` are the doubled (4.12) exactly, zero class included. **No weakened statement was found.**

Deviations, all recorded in docstrings:

1. `eq_6_14` is stated with $I(\rho)$ and M_0 as *defined real numbers* — the closed form (A.2) and $-1329/200$ — not as the logarithmic energy. Do not read “(6.4) is proved” as “Lemma 6.1 is proved”.
2. `lemma_3_1` proves (3.4) for every *partial* sum $\sum_{j \leq J} p^j U_j$, not for the series in the Tate algebra. It is also *not in the cone*: Proposition 4.1 goes around it (item 7). The same holds for `lemma_4_2` in its abstract-field form; Proposition 4.3 uses the rank form `vanishing_of_rank`.
3. `outer_local_analysis` states the **literal rank form of (4.10)**, $\text{rank } L \leq r_p$. An earlier formulation of this interface statement, as a set of vanishing columns of L in the basis (4.11), was false for the paper’s own L (found in the referee audit of the preprint) and was replaced before the proof was written. The proof shows why: the vanishing is in the monomial basis, exactly as p. 12 argues, and $L = UL_0U^\top$ carries the rank bound over. The statement proved is (4.10) as printed.
4. `AppendixB.eq_5_16_5_17` proves the finite-interval difference form rather than the two improper integrals (5.16) and (5.17) — formally weaker than p. 16, and exactly what is consumed.
5. **Lemma 3.3 is proved by a different route.** The polynomial part does not use the paper’s two-scale estimate $P(\mathbb{Z}_p) \subset p^{-L_0-M_0}\mathbb{Z}_p$, (3.8) or (3.9); it expands A in the binomial basis $\binom{x+K}{k}$ and computes τ of each quotient exactly. The *statement* of Lemma 3.3 is verified; its printed proof is not checked line by line. The $-v_p(24)$ turns out to be unnecessary.
6. `eq_5_7_uniformity` bundles three claims with one constant, proved with $C = 400M^2$.
7. **Lemma 3.2 (3.7) is not formalised as stated.** The Tate-algebra extension, (3.5), the parameter $Y = p^5X + C_p$, and (3.7) for rational functions do not appear. Proposition 4.1 is proved without them: pole terms bounded directly, the polynomial part through Raabe (the polynomial half of (3.7), the paper’s own justification) and a truncated-inverse estimate. Do not say “Lemma 3.2 is formalised”.

8. $b_a < 6\alpha x + 3$ strictly, where the paper prints $\leq$ (towards strength; see item 18).
9. Lemma 4.2 takes $v(\text{pinv}) \geq -1$ rather than literally p^{-1} (more general).
10. (5.21) and (5.16)–(5.18) are stated for $M \in 40\mathbb{Z}_{>0}$, as printed.
11. (5.8)/(5.9) are nested `ifs`, differing from the paper on a null set.
12. Naming trap: `InnerAlloc.L 0` is not the paper’s $L_0 = 4M + 10$.
13. (6.11) is proved as printed via a sharper intermediate bound.
14. `PrimeSum.RR_bddOn` proves the crude $|R| \leq 200M^2$, not the paper’s sharp bound; nothing needs more.
15. **The p. 12 divided-difference step needs more than the paper says.** $H_{p-a}^{(5)} \equiv H_{a-1}^{(5)} \pmod{p}$ alone leaves a residue $-1/a$ (machine-checked); the proof also uses the $-1/4 + 1/(2j)$ tail of (2.3), and the must-fail control shows it is essential. A point the author may want to revise, not an error in the result.
16. Unimodularity of (4.5) and (4.11) is proved by one mod- p Hermite-basis argument rather than the paper’s two separate justifications.
17. The §4.2 local analysis uses only $p \geq 7$, $K < 3p$, $2K < p^2$, $2N < p$, $5N \leq 2p - 2$ — neither $p \leq K$ nor $K \geq 200M^2$ (towards strength).
18. **A gap on p. 10, repaired.** The paper asserts $b_a \leq 6\alpha x + 3$ without proof, attributing it to (4.4); it is a statement about ℓ_N , equivalent to $p\ell_N(a) \leq 2N + p$. The naive bound $\ell_A(a) \leq 2\lfloor A/p \rfloor + 2$ gives only $b_a \leq 6\alpha x + 6$, which does not make the dimensions of (4.4) provably nonnegative. The inequality is proved in the strict form $p\ell_A(a) < 2A + p$ by the case split on $v_A = A \bmod p$ (`Zeta5.el11_lt_caseSplit`) and independently from the closed form of ℓ_A (`Zeta5.el11_lt`, the one in the dependency cone); Lean checks by `rfl` that the two statements are identical. A point the author may want to revise, not an error in the result.

7 Trusting the audit metaprogram

`Zeta5/Audit.lean` walks the dependency cone and reports the groups above. It avoids two traps that a naive dependency walker falls into (`ConstantInfo.value?` is `none` for theorems in Lean 4.34; reachability must match `Lean.CollectAxioms.collect`), checks itself against `Lean.collectAxioms` on every build, and checks that no `sorry` escapes the cone. Its `#sorry_tree` controls for the four interface statements of the previous section, the shared lemma and seven of the main new theorems all print `depends on NO sorry`, and `eq_6_14` prints `rests on 1 sorry(s)`. An adversarial certification (`docs/CERTIFICATION.md`; scripts in `cert/`, sharing no code with `Audit.lean`) agrees: 62097 constants in the cone, one `sorry`, two project axioms, zero orphans, a full kernel replay of every module by `leanchecker`, and the deliberately broken `value?`-walker reporting the wrong answer (0 sorries), as a negative control should.

8 What to do next

(1) `RealBound.eq_6_14`, the only `sorry`, in its three pieces (Andréief; Lemma 6.2 and (6.6)–(6.9); (6.2) via Appendix A), each decomposed into numerically tested lemma statements first.

(2) Reduce `pnt_prime_riemann_sum` to $\theta(x) \sim x$ (Abel summation plus a Darboux sandwich).
 (3) Formalise p. 5’s four integrations by parts, so that `hermite_pole_integral` becomes DLMF 25.11.29 verbatim. (4) Points the author may want to revise: the p. 10 inequality (item 18), the p. 12 divided-difference step (item 15), and the phrasing of the rank argument for (4.10) (item 3).

How to report this result. “Theorem 1.1 is machine-checked conditional on one explicitly named unproved statement of the paper — (6.14), the logarithmic-energy bound of §6 — plus Hermite’s integral formula and the prime number theorem in partial-summation form.” Never “modulo standard facts”: (6.14) is Fauzan’s own claim, and it is the analytic heart of §6.